IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE/EIE PART II: MEng, BEng and ACGI

DIGITAL ELECTRONICS 2

Friday, 5 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 50% of the marks. Questions 2 and 3 carry equal marks (25% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.Y.K. Cheung
Second Marker(s) : W. Dai

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Information for Candidates:

The following notation is used in this paper:

1. Unless explicitly indicated otherwise, digital circuits are drawn with their inputs on the left and their outputs on the right.
2. Within a circuit, signals with the same name are connected together even if no connection is shown explicitly.
3. The notation X[2:0] denotes the three-bit number X2, X1 and X0. The least significant bit of a binary number is always designated bit 0.
4. Signed binary numbers use 2's complement notation.

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1. (a) The circuit in *Figure 1.1* has two flip-flops FF1 and FF2 with a clock to Q delay of t_{CQ}, setup time of t_s and hold time of t_h. The combinational logic block A between the two flip-flops is implemented with an ALM from an FPGA, which has a propagation delay of t_A. The clock signal to the second flip-flop is driven by the symmetrical clock signal CLK via inverter B with a propagation delay of t_B.

(i) Write down the setup and hold inequalities that apply to the flip-flop FF2.

[5]

(ii) Calculate the maximum frequency of the clock signal CLK for reliable operation if $t_S = 3$, $t_H = 2$, $3 \leq t_{CQ} \leq 4$, $1 \leq t_A \leq 3$ and $2 \leq t_B \leq 8$, where all times are in nanoseconds.

[5]

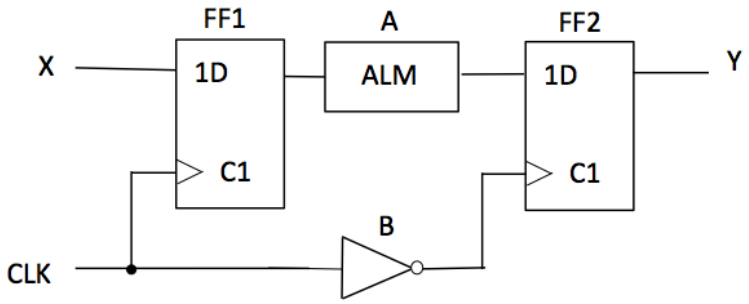


Figure 1.1

(b) *Figure 1.2* shows part of a microcontroller system with an 8-bit data bus and a 16-bit address bus interfacing to a 16 kilobyte read only memory (ROM).

(i) State how many address bits the ROM will require.

[2]

(ii) The ROM occupies the memory space in hexadecimal 16'h8000 to 16'hBFFF. Derive the Boolean express for the chip enable signal CE such that the ROM is only enabled when CLOCK and RD are both high.

[4]

(iii) If the ROM is to be implemented using two 8k x 8 Flash memory chips, draw a diagram showing how these two Flash memory chips can be interfaced to the microcontroller.

[4]

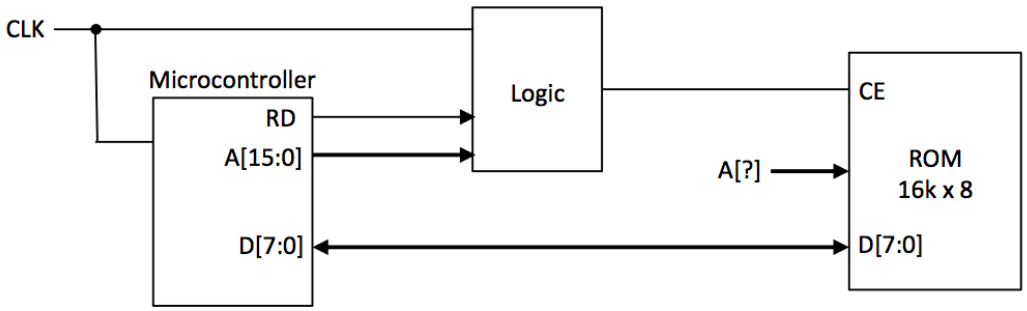


Figure 1.2

(c) *Figure 1.3(a)* shows a Finite State Machine (FSM) with two state bits Q1 and Q0, an input A, an output X and a clock signal CLK. The circuit consists of two flip-flops, a NAND gate and an XOR gate. The initial state of the FSM is 0.

(i) Draw the state diagram for this circuit.

[3]

(ii) Hence or otherwise, complete the timing diagram shown in Figure 1.3(b) showing the state of the FSM and the waveform for the output X.

[3]

(iii) Implement this FSM in Verilog HDL.

[4]

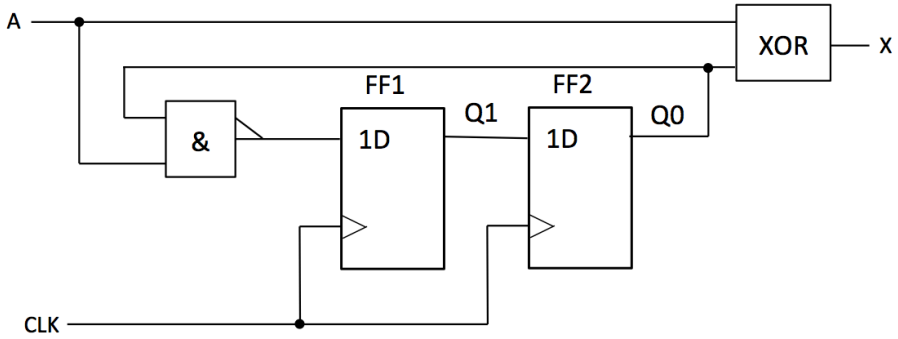


Figure 1.3(a)

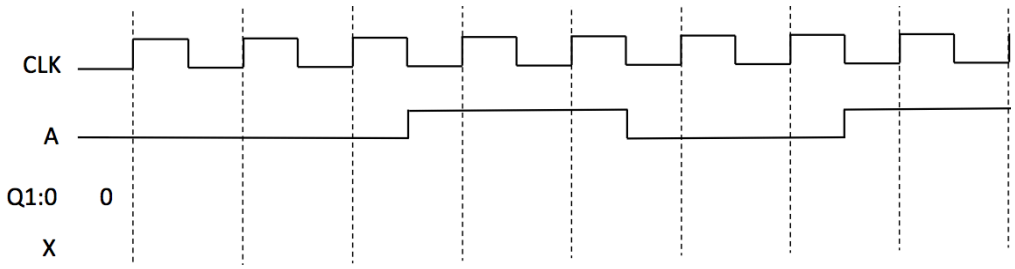


Figure 1.3(b)

(d) *Figure 1.4* shows a circuit with five flip-flops FF1 to FF5, two XOR gates and a 2-to-1 multiplexer MUX. The initial states of the flip-flops are all set to one.

(i) Assuming that S = 0, lists the values of Q5:1 for 5 clock cycles. (You may use binary or hexadecimal representations.)

[3]

(ii) Shortly after the rising edge of the CLK at the 5th clock cycle, S changes to 1. What are the values of Q5:1 for the next 5 clock cycles?

[3]

(iii) Specify this circuit in Verilog HDL.

[4]

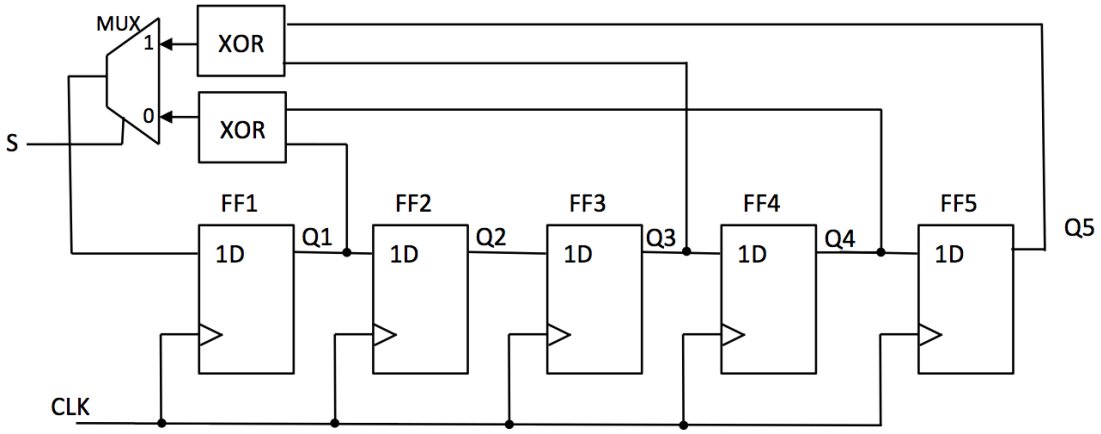


Figure 1.4

(e) *Figure 1.5* shows a 3-bit digital-to-analogue converter (DAC) with a 3-bit digital input *X*[2:0], each bit controlling the electronic switches where a 0 corresponds to the down position. V_R is a reference voltage source at 2V.

(i) Explain why the current I0 is independent of value X0.

[3]

(ii) Determine the value of I_0, I_1 and I_2.

[3]

(iii) Calculate the value of V_{OUT} when the input *X*[2:0] = 5.

[4]

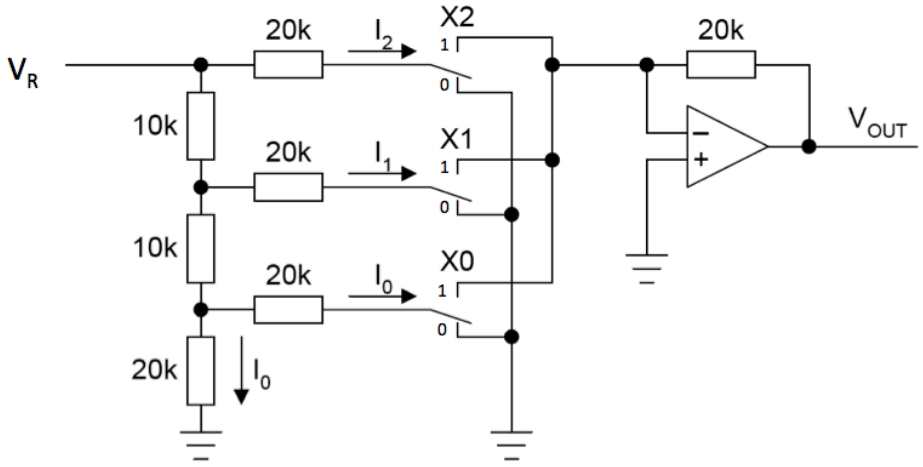


Figure 1.5

2. *Figure 2.1 (a)* shows a digital circuit that computes the approximate square root of an 8-bit number *P*[7:0] and produces a 4-bit answer *X*[3:0]. The circuit consists of a squaring circuit, a digital comparator and a finite state machine that implements the successive approximation algorithm. The digital comparator output *HIGHER* is high if the value at its input *A* is larger than its input *B*. The FSM is controlled by the clock signal *CLK*. It starts the computation when *LOAD* goes from low to high and the *DONE* signal then goes low until the computation is completed and *X*[3:0] contains the valid result.

(a) Assuming that *P*[7:0] = 8'h54, complete the timing diagram shown in *figure 2.1 (b)*.

[7]

(b) Hence or otherwise, explain how this circuit is able to computer the square root of the input. State any assumptions used.

[8]

(c) Design in Verilog HDL the finite-state machine SAR FSM.

[10]

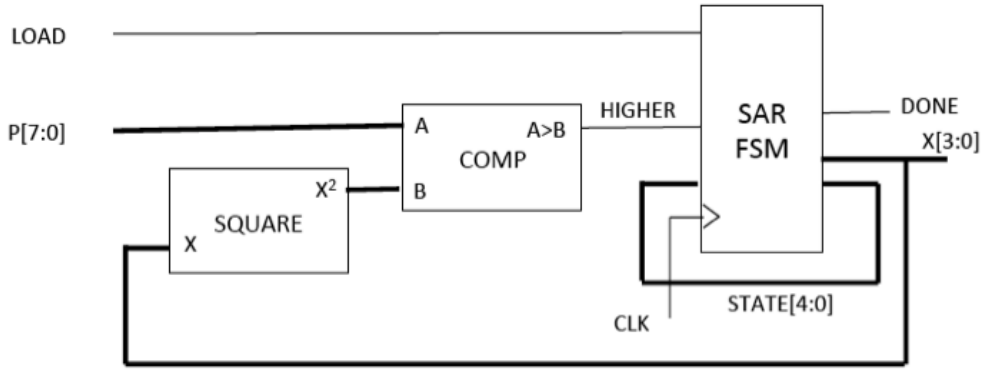


Figure 2.1 (a)

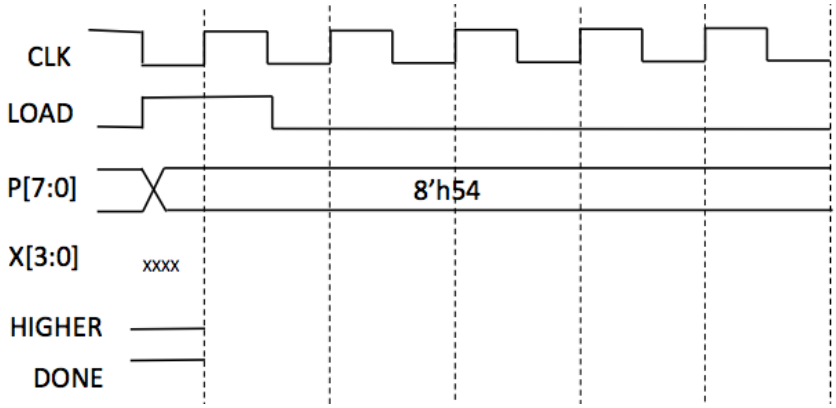


Figure 2.1 (b)

3. *Figure 3.1* shows a circuit consisting of a 2-bit up counter CTR2, a 3-bit counter CTR3 which counts up when EN is high, a 2-input AND gate, a 3-bit register FF which is enabled when G2 is high and a 3-bit binary subtractor SUB. The circuit is clocked by the signal CLK, and has an input A and an output Z[2:0].

(a) Specify this circuit in Verilog HDL.

[10]

(b) Assuming that both counters and the register are initially reset, complete the timing diagram in *figure 3.2* by showing the values of K[1:0], Q[2:0], Y[2:0] and Z[2:0].

[15]

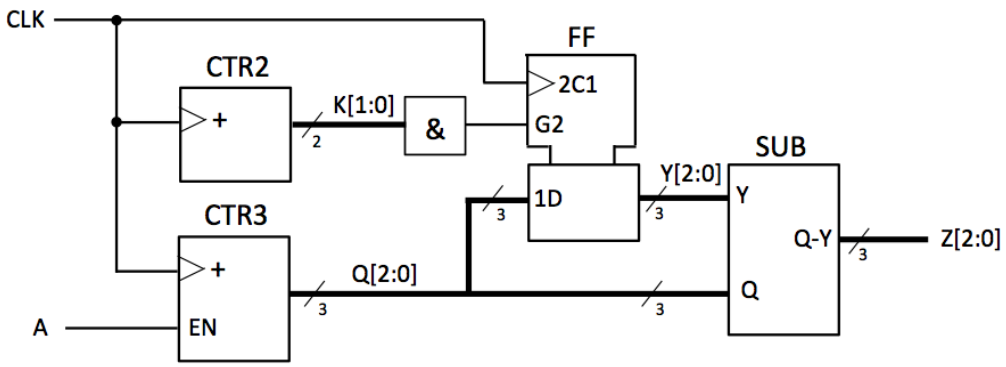


Figure 3.1

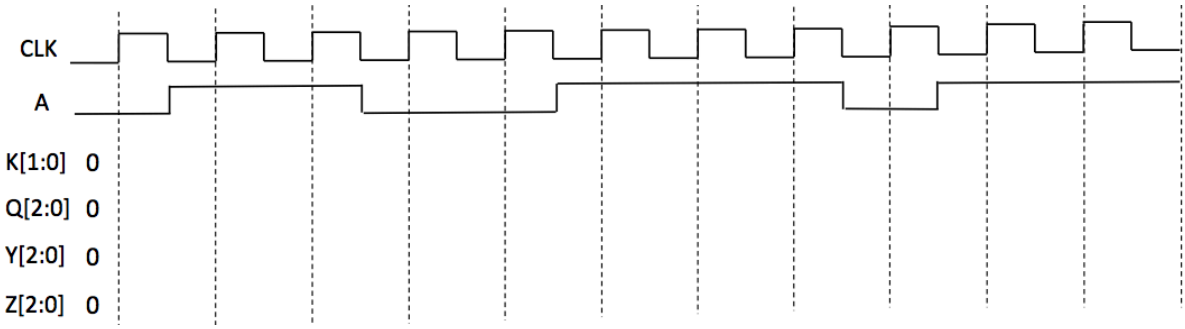


Figure 3.2